Module 7 Assignment 7A:

Hello Route

Course: CS 85 - PHP Programming
Instructor: Vicky Seno

For this assignment, I created a standalone Laravel application that demonstrates framework installation, web routing, dynamic route parameters, testing, and Git-based project delivery. The project also runs inside my larger CS85 coursework application, so it works as an app inside an app while remaining connected to the Module 7 roadmap.

The required /hello route returns a fixed message, while /greet/{name} reads a value from the URL and creates a personalized response. I extended the required version so that hyphenated names such as serge-hall are displayed as Serge Hall. I also added an optional instructor spotlight for Vicky Seno as a non-standard example of how the same route can return a designed Blade view instead of plain text.

Assignment URL: http://127.0.0.1:8000/assignments/module7a

Standalone URL: http://127.0.0.1:8007

GitHub Repository: https://github.com/sergehall/cs85-php-programming/tree/main/assignments/module7a

Project Summary

- Laravel project created with the official Laravel installer.
- A static /hello route and a parameterized /greet/{name} route.
- Feature tests for normal, hyphenated, instructor, and invalid route values.
- A responsive assignment page and a separate Vicky Seno presentation view.
- Ten logical Git commits that document the development process.

PART 1 · LARAVEL INSTALLATION & PROJECT SETUP

Step 1: Verify Prerequisites

Before creating the project, I verified that PHP and Composer met the assignment requirements. The assignment requires PHP 8.2 or higher and Composer 2.x or higher. My system exceeded both minimum versions.

```
php --version
composer --version
laravel --version
```

Verified environment: PHP 8.5.8, Composer 2.10.2, and Laravel Installer 5.25.1.

Screenshot 1

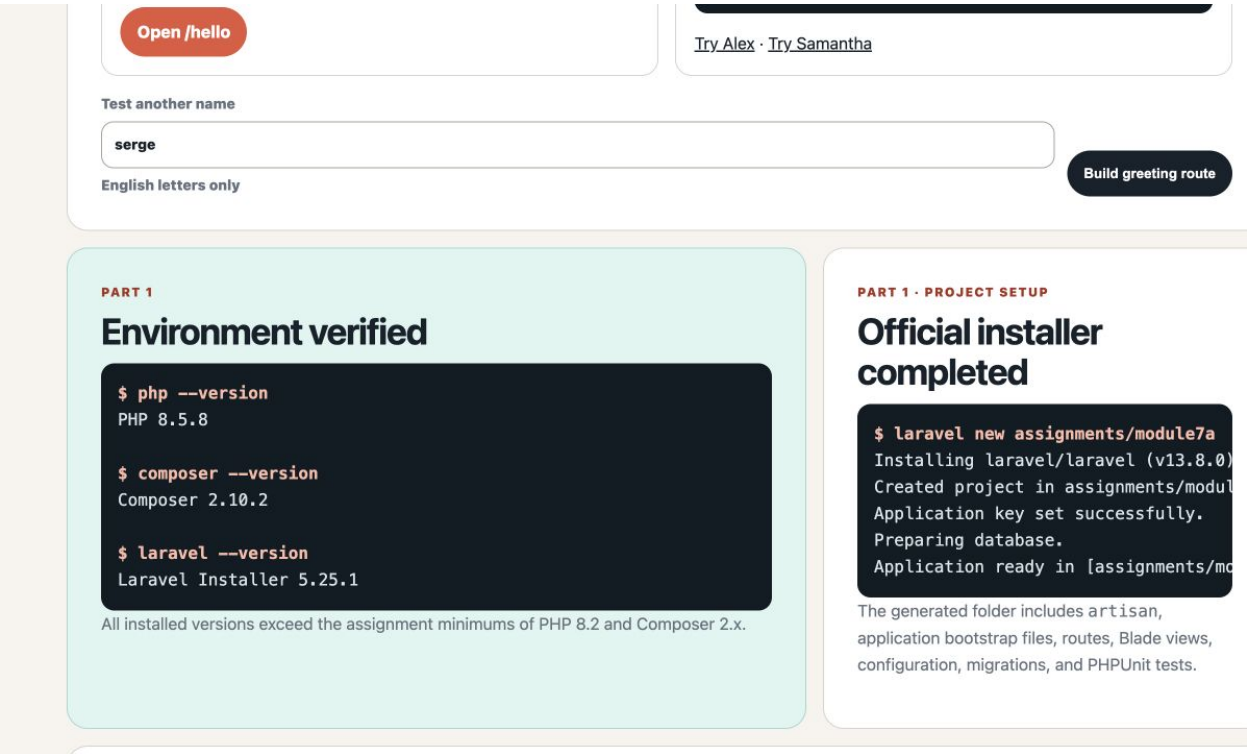


Figure 1. Terminal output showing the installed PHP, Composer, and Laravel Installer versions.

Step 2: Install Laravel and Create the Project

I installed the Laravel command-line installer with Composer and generated the assignment as a complete Laravel application inside the course repository.

```
composer global require laravel/installer
laravel new assignments/module7a --no-interaction \
    --phpunit --no-boost --database=sqlite
cd assignments/module7a
```

Screenshot 2

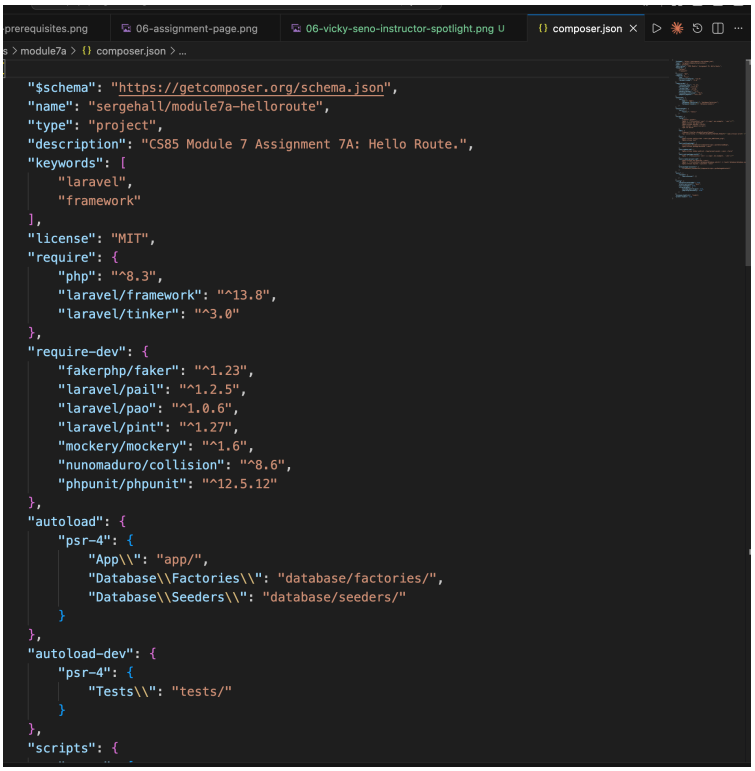


Figure 2. Module 7A composer.json showing the Laravel framework dependency and project configuration.

Run the Standalone Application

```
cd assignments/module7a
php artisan serve --port=8007
```

The local .env file stores machine-specific environment settings and is excluded from Git. The safe .env.example file remains in the repository so another developer can configure a fresh clone without exposing secrets.

PART 2 · LARAVEL CONCEPTS TUTORIAL

What Is Laravel?

Laravel is a modern PHP framework for building web applications. A framework provides an organized structure and reusable tools so a developer does not need to recreate routing, validation, database access, authentication, security features, and testing support for every new project.

Laravel is useful because its conventions make a project easier to read and maintain. The same folder structure and naming patterns are used across many Laravel applications, which helps developers understand unfamiliar code more quickly.

What Is a Route?

A route connects an HTTP method and URL to the code that should run. When a browser requests a page, Laravel compares the URL with the routes defined in routes/web.php. If a matching route is found, Laravel runs its closure or controller method and sends the response back to the browser.

Important Laravel Files and Folders

- routes/web.php contains browser-facing route definitions.
- resources/views contains Blade templates used to generate HTML pages.
- app/Http/Controllers contains controllers for request-handling logic that grows beyond a small closure.
- .env contains local environment values and secrets and must not be committed.
- tests/Feature contains tests that verify complete HTTP behavior.

Request Flow

1. The browser sends a GET request to a URL such as /greet/alex.
2. Laravel matches the request to the route in routes/web.php.
3. The route receives alex as the $name parameter and formats the response.
4. Laravel sends Hello, Alex! back to the browser.

PART 3 · DEFINE TWO ROUTES AND PUSH TO GITHUB

Route 1: Static /hello Route

The first route is static because it always returns the same response. It does not need input from the URL.

```
Route::get('/hello', function () {
    return 'Hello from Laravel!';
});
```

Test URL: http://127.0.0.1:8000/assignments/module7a/hello

Screenshot 3

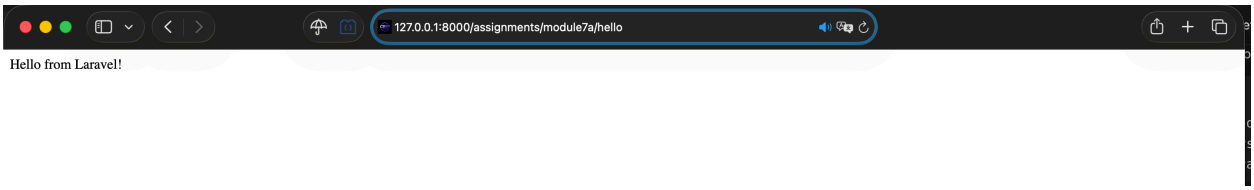


Figure 3. Browser output showing that /hello returns Hello from Laravel!

This confirms that Laravel matched the GET request and returned the exact message required by the assignment.

Route 2: Dynamic /greet/{name} Route

The second route contains the dynamic {name} segment. Laravel captures that part of the URL and passes it to the route function as $name. My implementation also supports first and last names separated by a hyphen.

```
Route::get('/greet/{name}', function (string $name) {
    $displayName = str_replace(
        '-', ' ', ucwords(strtolower($name), '-')
    );

    return 'Hello, '.$displayName.'!';
})->where('name', '[A-Za-z]+(?:-[A-Za-z]+)*');
```

Screenshot 4

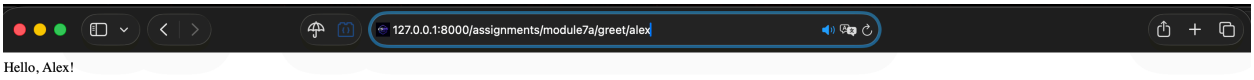


Figure 4. Browser output showing that /greet/alex returns Hello, Alex!

Screenshot 5

Hello, Samantha!

Figure 5. Browser output showing that /greet/samantha returns Hello, Samantha!

Non-Standard Enhancement: Vicky Seno Instructor Spotlight

As an additional example beyond the assignment requirements, I created a Vicky Seno button on the assignment page. It still uses the same /greet/{name} route pattern. The URL value vicky-seno is converted into the display name Vicky Seno, but this specific value returns a dedicated Blade presentation instead of a plain-text response.

Instructor Spotlight URL: http://127.0.0.1:8000/assignments/module7a/greet/vicky-seno

Screenshot 6

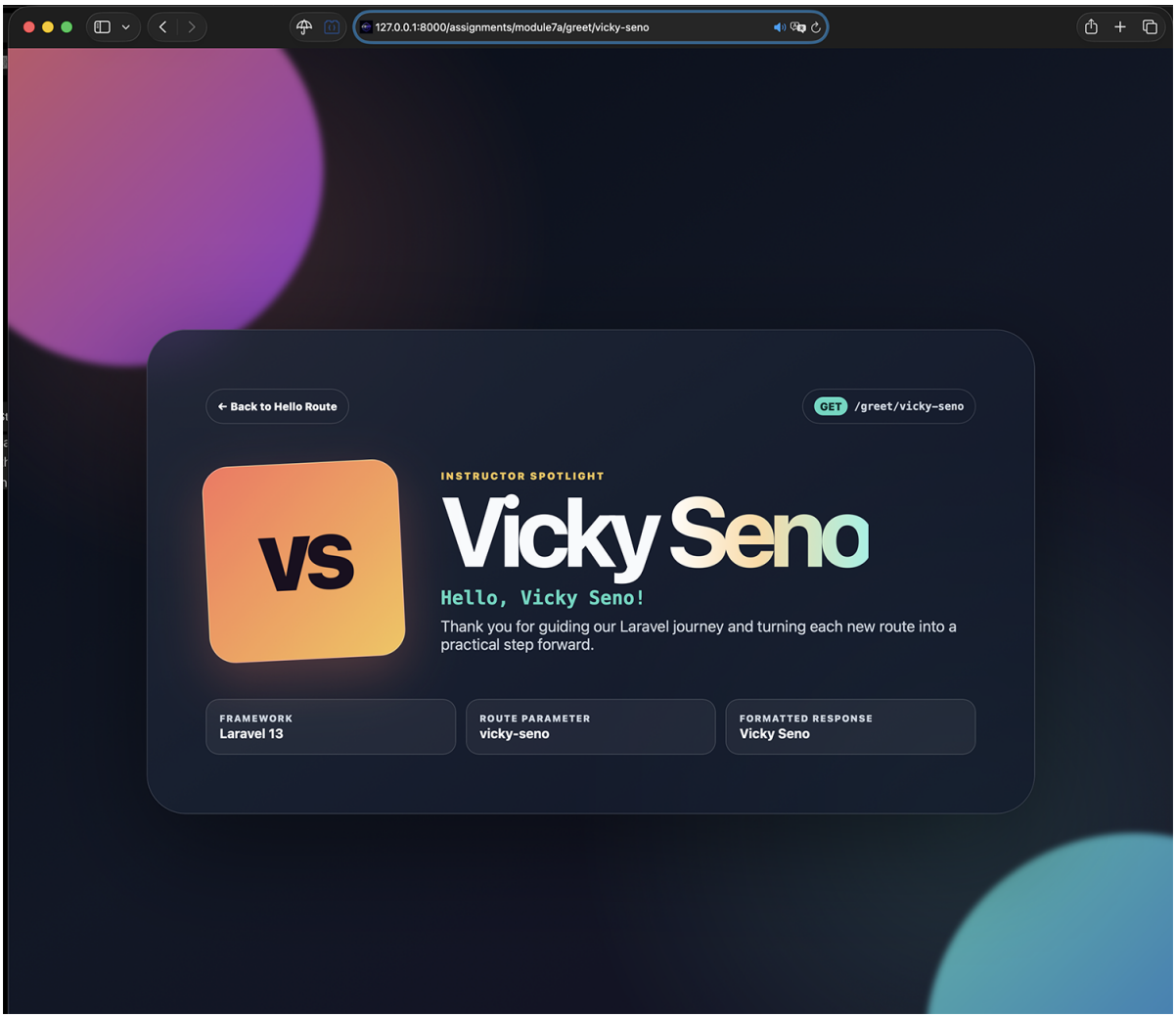


Figure 6. Result displayed after clicking the Vicky Seno button on the assignment page.

The presentation displays the GET route, an Instructor Spotlight label, a VS monogram, the formatted greeting Hello, Vicky Seno!, and a short thank-you message. CSS reveal, floating, orbit, and sheen animations create a presentation-style entrance. A prefers-reduced-motion rule disables unnecessary motion for users who request a reduced-motion experience.

This enhancement demonstrates that one parameterized Laravel route can choose between a simple string response and a complete Blade view while preserving the same routing concept.

Testing and Version Control

Feature Testing

I added feature tests so the required behavior can be verified automatically instead of relying only on browser checks. The standalone Module 7A suite contains six passing tests with twenty-six assertions. The integrated CS85 application contains 113 passing tests with 1,015 assertions.

```
cd assignments/module7a
php artisan test

# From the main CS85 project
php artisan test
```

- The landing page explains both required routes.
- /hello returns the exact required text.
- Single and hyphenated names are capitalized and formatted correctly.
- The Vicky Seno value returns the instructor presentation.
- Spaces, repeated hyphens, digits, and unsafe characters are rejected.

Git and GitHub

I organized the completed work into ten logical Git commits. The history shows the Laravel scaffold, required routes, interactive page, responsive styling, instructor spotlight, tests, README, screenshot evidence, course integration, and quality checks as separate development steps.

```
git init
git add .
git commit -m "Initial commit: Laravel routes assignment"
git remote add origin https://github.com/sergehall/module7a-helloroute.git
git push -u origin main
```

The current integrated repository location is https://github.com/sergehall/cs85-php-programming/tree/main/assignments/module7a.

PART 4 · REQUIRED KNOWLEDGE CHECK

What this assignment demonstrates

What is Laravel and why is it useful for developers?

Laravel is a PHP framework that gives developers an organized way to build web applications. It provides a standard project structure and built-in tools for common tasks such as routing, form validation, database access, authentication, security, and automated testing. Without a framework, a developer would need to design and maintain many of these parts from the beginning for every project.

Laravel is useful because it lets developers focus more attention on the purpose of an application instead of repeatedly solving the same setup problems. Its clear conventions also make code easier to read, debug, and share with other developers. In this assignment, I could define a working URL and response in only a few lines while Laravel handled the request and response process behind the route.

What does the /greet/{name} route do?

The /greet/{name} route reads the value placed after /greet/ in the URL. Laravel stores that dynamic value in the $name parameter and passes it into the route function. The function formats the value and combines it with the word Hello to create a personalized response. For example, /greet/alex returns Hello, Alex!.

My version also supports a first and last name separated by a hyphen because a normal space should not be typed directly inside this route segment. The URL /greet/serge-hall is converted into Hello, Serge Hall!. A route constraint allows English letters and correctly placed hyphens while rejecting values such as serge--hall, names containing digits, and unsafe characters.

Why is it important to use Git and GitHub in software projects?

Git records a history of meaningful changes made during development. Each commit acts like a saved checkpoint that explains what was added or corrected. If a new change creates a problem, a developer can compare versions, find where the problem began, and return to an earlier working state when necessary. A clear commit history also makes a project easier to review because a teacher or teammate can follow the development process instead of seeing only the final files.

GitHub stores the Git repository online and provides a remote backup of the project. It also makes the work easy to share, review, and continue from another computer. In professional software projects, GitHub supports collaboration through branches, pull requests, issue tracking, automated checks, and code review. For this assignment, it provides evidence that the Laravel project and its history were published successfully.

How is this project organized?

The assignments/module7a folder is a complete standalone Laravel application. It has its own Artisan command, Composer dependencies, configuration, web routes, Blade views, tests, and local environment file. It can run independently on port 8007, which directly demonstrates the project structure requested by the assignment.

The main CS85 Laravel project also mounts Module 7A at /assignments/module7a and connects it to the Module 7 roadmap. This app-inside-an-app organization keeps all coursework in one repository while preserving a real Laravel application inside the assignment folder. It also lets the same route examples be presented as part of the larger course website.

PART 5 · Completed Assignment Page

The completed Module 7A assignment page presents the required /hello and /greet/{name} routes in an interactive format. It includes route explanations, example responses, testing controls, Laravel project information, and the required knowledge-check answers.

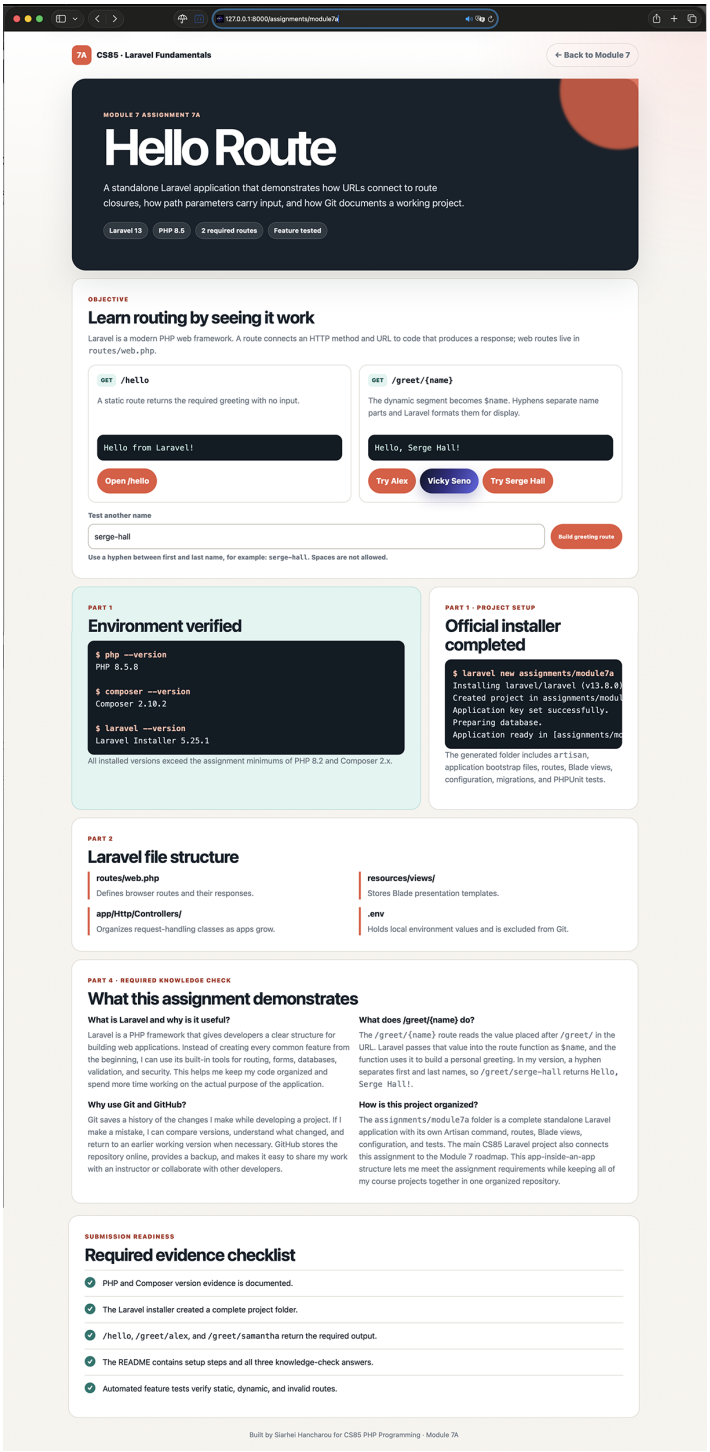


Figure 6. Completed Module 7A assignment page showing the required Laravel routes and interactive testing controls.